

Review

Data-centric, robust, and explainable multimodal deep learning for clinical decision support: A systematic review

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HIGHLIGHTS

- Reviewed 31 recent multimodal, data-centric clinical decision support studies.
- Data-centric and explainability techniques were reported in ~74%–84% of studies.
- External validation was conducted in only 13%; a clinical outcome was measured in only 10%.
- 87% of the studies were at high overall risk of bias (PROBAST + AI).
- Proof-of-concept, not proof-of-benefit: external validation is the priority.

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ABSTRACT

Purpose: Multimodal deep learning is increasingly proposed for clinical decision support (CDS) under a “data-centric” framing that prioritizes label quality, missing-modality robustness, distribution shift, calibration, and explainability. Prior reviews have examined multimodal medical AI, CDS, and data-centric methods separately, but none address their intersection. We mapped the modalities, fusion strategies, and data-centric and explainability techniques used in this recent literature, quantified how often each is implemented rather than merely mentioned, assessed deployment-relevant evidence (external validation, clinical-outcome measurement, equity), and formally appraised study-level risk of bias.

Methods: Following the PRISMA 2020 statement (PROSPERO CRD420261427815; registered retrospectively), we screened 150 records and included primary, clinical, multimodal studies that applied machine or deep learning to a decision-support task and reported at least one quantitative result. Two reviewers screened and extracted data with consensus adjudication. Each study was coded against pre-specified operational definitions, separating implemented or empirically evaluated techniques from those only mentioned. Study-level risk of bias was assessed with PROBAST + AI. Synthesis was narrative.

Results: Thirty-one studies met inclusion; 30 (97%) were published between 2024 and 2026, with a median of three modalities (range 2–6), most commonly structured EHR (71%) and imaging (39%). Data-centric techniques were frequently reported (74–84% across label-noise, distribution-shift, calibration, missing-modality and class-imbalance handling; equity 61%). However, external validation was reported in only 4/31 studies (13%), a clinical or provider outcome in 3/31 (10%), and no study reported routine deployment. Overall risk of bias was high in 27/31 studies (87%), driven by the analysis domain.

Conclusion: Within this recent, self-selected slice of the field, technical robustness and explainability techniques are widely reported but rarely validated out-of-distribution or against clinical outcomes, and the underlying evidence is at high risk of bias. Progress requires external multi-site validation, clinical-outcome measurement, formal bias appraisal, and adherence to AI reporting standards (e.g., TRIPOD + AI) before deployment can be justified.

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1386-5056/© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

1. Introduction

Clinical reasoning integrates heterogeneous evidence—medical imaging, structured electronic health records (EHRs), physiological signals, clinical text, and increasingly genomic and wearable data. Multimodal deep learning, which models these streams jointly, is an intuitively attractive substrate for clinical decision support (CDS), and reported gains over unimodal baselines have driven rapid uptake [1,2]. A recent scoping review of multimodal medicine maps the same expansion across imaging, text, and structured data [3]. In parallel, methodological attention has shifted from architecture alone toward a data-centric view, in which label quality, missing or asynchronous modalities, class imbalance, distribution shift, and predictive uncertainty are treated as first-class design problems rather than preprocessing afterthoughts [4,5]. Large language models are increasingly folded into such pipelines for clinical reasoning, though their reliability in real decision-making remains under scrutiny [6]. Distribution shift in particular is a recurring threat to transportability [7]. Explainability has become a near-mandatory companion, motivated by clinician trust, accountability, and emerging regulatory expectations [8,9].

These three threads—multimodality, data-centric robustness, and explainability—are increasingly invoked together as the headline contribution of a study. Yet the presence of a robustness *technique* does not establish that a model is reliable in deployment. Internal validation routinely overstates real-world performance, and robustness demonstrated on a single development dataset may not survive the distribution shifts introduced by new sites, devices, and populations [10]. Reporting and appraisal standards for clinical prediction models—TRIPOD + AI, DECIDE-AI, and PROBAST + AI—exist precisely to separate demonstrated benefit from technical promise [11,12], and are complemented by Good Machine Learning Practice and model-appraisal tools [13,14]. The challenge of connecting predictive performance to clinical implementation is well illustrated in specific domains such as heart-failure detection, where biomarker- and AI-based prediction still require prospective validation before clinical use [15]. Recent systematic reviews in adjacent clinical-AI domains have repeatedly found favorable model metrics coexisting with absent external validation and unmeasured patient benefit [16,17].

No systematic review has yet examined the specific intersection of data-centric robustness, explainability, and multimodality in CDS, or asked whether the robustness this literature claims is actually validated and at what risk of bias. We address that gap for the recent (2021–2026) “data-centric” literature. We pose three research questions:

- (RQ1) Which modalities, fusion strategies, and data-centric and explainability techniques do primary multimodal CDS studies employ?
- (RQ2) How often is each technique implemented or evaluated rather than merely mentioned?
- (RQ3) To what extent do these studies provide deployment-relevant evidence – external validation, measured clinical outcomes, equity analysis, standardized reporting – and at what risk of bias?

To our knowledge, this is the first review to synthesize this intersection, to apply an explicit implementation-versus-mention coding lens, and to pair it with a formal PROBAST + AI risk-of-bias appraisal.

2. Methods

2.1. Protocol and registration

This review was conducted and reported in accordance with the PRISMA 2020 statement [18]. The review was registered with PROSPERO (CRD420261427815) [19]. Registration was retrospective: it was completed after the searches and synthesis had been carried out. We disclose this here for transparency, and the registered record was aligned with the methods actually used. The completed PRISMA 2020 checklist is provided as Supplementary Material.

2.2. Eligibility criteria

Studies were eligible if they (i) were primary studies developing or evaluating a model or system—not reviews, surveys, editorials, commentaries, protocols, books, or bibliometric papers; (ii) were multimodal, combining two or more distinct data modalities (imaging, structured EHR/tabular, physiological signals/time-series, clinical text, genomics/omics, or wearable/sensor) within a single model or system; (iii) applied machine or deep learning to a clinical decision-support task (diagnosis, prognosis, risk stratification, triage, deterioration, or monitoring) in humans; and (iv) reported at least one quantitative result. Non-clinical applications, single-modality studies, purely conceptual or qualitative works without quantitative evaluation, and secondary literature were excluded. Preprints and theses were eligible only if they met all criteria, and their inclusion was tested in a sensitivity analysis (Section 3.7).

2.3. Information sources and search strategy

A search was conducted across four bibliographic databases (PubMed, Scopus, IEEE Xplore, and Web of Science), combining three concept blocks – multimodal/data-centric deep learning; robustness and explainability (missing modality, label noise, distribution shift, calibration, interpretability); and clinical decision support – using controlled-vocabulary and free-text terms joined with Boolean operators. The search was limited to 1 January 2021–2026 and to English-language records. The complete, database-specific search strings and final search dates are provided in Supplementary Material (S3). No grey-literature or citation-chaining searches were performed; this is acknowledged as a limitation. The four databases returned 365 records in total; after removal of 215 duplicates, 150 unique records remained for screening.

2.4. Study selection and data extraction

Records were consolidated in a reference manager and de-duplicated by matching on title, authors, and DOI, followed by a manual check of near-duplicates; 215 duplicate records were removed, leaving 150 unique records. Two reviewers independently screened titles/abstracts and then full texts against the eligibility criteria; disagreements were resolved by discussion, with a third reviewer consulted where consensus could not be reached. Two reviewers independently extracted data onto a standardized form, and discrepancies were reconciled by consensus. Screening decisions, and the list of full-text records excluded with reasons, are provided in Supplementary Material (S8).

2.5. Coding framework and operational definitions

Each study was coded for bibliographic data, clinical task and setting, modalities and modality count, dataset provenance (public/institutional/synthetic), sample size, AI paradigm, fusion strategy, five data-centric techniques, explainability method, validation strategy, external validation, primary metric, deployment status, measured clinical/provider outcome, equity/subgroup analysis, and reporting standard. To ensure reproducibility, each coded item used a pre-specified operational definition (full codebook in Supplementary Material S5); the core definitions were: *Implemented/evaluated* – the technique was built into the model or empirically tested with a reported result (coded *present*); *mentioned* – the technique was named only as a challenge, gap, or future direction (coded *absent*). *Missing-modality handling* – an explicit mechanism (imputation, dropout, shared-representation, or generative reconstruction) evaluated under absent modalities. *Distribution-shift handling* – domain adaptation, external/temporal evaluation, or shift-detection with a reported effect. *Equity analysis* – reported model performance disaggregated by at least one demographic or clinical subgroup. *Clinical outcome* – a measured patient, clinician, or health-system outcome (e.g., length-of-stay, decision accuracy in workflow, time-to-care), as distinct from a technical performance metric. This strict rule is

necessary because, in an emerging field, naming a problem is frequently mistaken for solving it.

2.6. Risk-of-bias assessment

Study-level risk of bias and applicability were assessed with PROBAST + AI [11,13], rating each study across the participants, predictors, outcome, and analysis domains as low, high, or unclear, with the overall judgement taken as the highest severity across domains. Two reviewers rated each study independently and reconciled their assessments by consensus. Domain-level results are summarized in Table 3 and the per-study traffic-light plot in Fig. 2; the full assessment is provided in Supplementary Material (S9).

2.7. Synthesis

Because designs and outcomes were heterogeneous, we used narrative synthesis with descriptive tabulation; no meta-analysis was performed. Proportions are reported as $n/31$ (%). To probe robustness of the findings, results were examined against the risk-of-bias appraisal and stratified where cell sizes permitted (Section 3.6), and a sensitivity analysis re-derived the headline proportions after excluding preprints and theses (Section 3.7).

3. Results

3.1. Study selection

Of 150 unique records screened, 119 were excluded and 31 primary, clinical, multimodal studies met all inclusion criteria and were synthesized (Fig. 1). Exclusions comprised secondary or non-primary literature (reviews, surveys, commentaries, books; the majority), single-modality studies, non-clinical applications, and studies without a quantitative clinical result. Of the 150 screened records, 111 were excluded at title/abstract (89 reviews or other secondary literature, 14 single-modality, 8 non-clinical) and 39 were assessed at full text, of which 8

were excluded (5 conceptual/commentary/book, 2 non-clinical, 1 qualitative without a quantitative result); the full list of full-text exclusions with reasons is provided in Supplementary Material (S8). Per-study characteristics, including dataset provenance, sample size where reported, study design, and whether any comparison against standard care was undertaken, are provided in Supplementary Table S1; no included study compared model outputs against standard clinical workflow, and sample size was inconsistently reported.

3.2. Study characteristics

The corpus is recent: 30/31 studies (97%) were published in 2024–2026 (2021, $n = 1$; 2024, $n = 7$; 2025, $n = 13$; 2026, $n = 10$), reflecting the rapid emergence of the data-centric multimodal framing. The median number of modalities was three (range 2–6); nine studies (29%) combined two modalities and three (10%) combined five or more. The most frequently used modalities were structured EHR/tabular data (22, 71%) and imaging (12, 39%), followed by physiological signals or time-series (10, 32%), wearable/sensor data (7, 23%), clinical text (6, 19%), and genomics/omics (5, 16%). Applications spanned diagnosis, risk stratification, deterioration, and longitudinal monitoring across oncology, neurology, cardiology/ICU, endocrinology, and other specialties.

3.3. AI paradigms, fusion, and explainability

Model families were dominated by deep and ensemble architectures (Table 1). Convolutional networks and ensemble/gradient-boosted models were most frequent (each 29%), followed by recurrent/LSTM (23%) and transformer/attention models (19%); large language or generative-AI components (13%) and federated-learning frameworks (16%) appeared in a minority, and several studies combined families. Early or feature-level fusion was most common (45%), followed by attention/intermediate fusion (32%) and late/decision-level fusion (13%); three studies (10%) did not specify a strategy. Explainability was near-universal but overwhelmingly post-hoc: SHAP (61%) and attention visualization (32%) predominated, followed by LIME (19%) and gradient/saliency maps (16%), with counterfactual and integrated-gradient methods rare (6% each). Formal evaluation of explanation faithfulness or stability was rare; explanations were typically presented as illustrative outputs rather than validated against ground-truth attributions, limiting their evidentiary value for clinician trust.

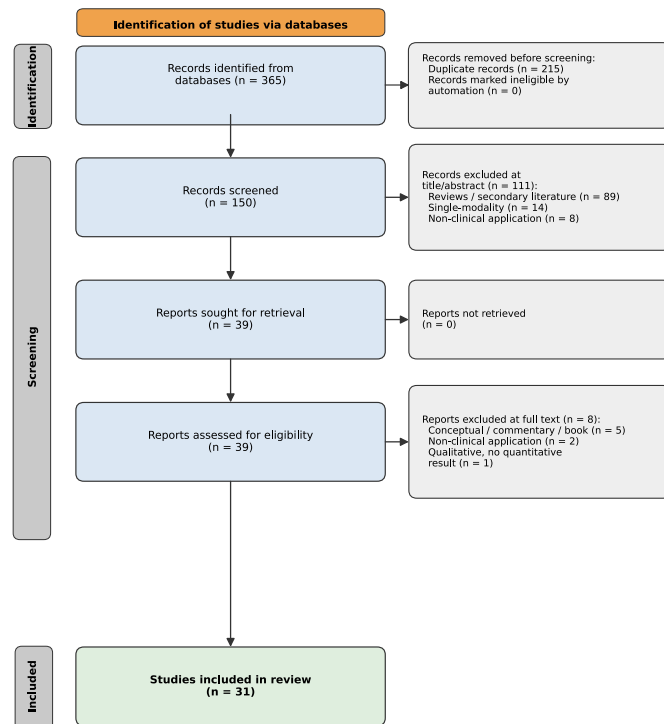


Fig. 1. PRISMA 2020 flow diagram of the study selection process.

Table 1
AI paradigm, fusion strategy, and explainability method (studies using each; $n = 31$). Categories are not mutually exclusive.

Category	Studies, n (%)
<i>AI paradigm / model family</i>	
Convolutional neural network	9 (29)
Ensemble / gradient boosting	9 (29)
Recurrent / LSTM	7 (23)
Transformer / attention	6 (19)
Federated learning	5 (16)
Large language / generative AI	4 (13)
<i>Fusion strategy</i>	
Early / feature-level	14 (45)
Attention / intermediate	10 (32)
Late / decision-level	4 (13)
Not specified	3 (10)
<i>Explainability method</i>	
SHAP	19 (61)
Attention visualization	10 (32)
LIME	6 (19)
Gradient / saliency maps	5 (16)
Counterfactual	2 (6)
Integrated gradients	2 (6)

Table 2
Data-centric techniques and deployment-readiness indicators (implemented or evaluated; $n = 31$).

Indicator	Studies, n (%)
<i>Data-centric techniques</i>	
Label-noise / data-quality handling	26 (84)
Distribution-shift handling	26 (84)
Calibration / uncertainty quantification	26 (84)
Missing-modality handling	25 (81)
Class-imbalance handling	23 (74)
Equity / subgroup analysis	19 (61)
<i>Deployment-readiness</i>	
External / multi-site validation reported	4 (13)
Patient / clinician / system outcome measured	3 (10)
Prediction-model reporting standard cited	<5
Prospective or in-workflow deployment	0 (0)

Table 3
PROBAST + AI risk-of-bias summary by domain (number of studies; $n = 31$).

Domain	Low	Unclear	High
Participants	14	12	5
Predictors	28	3	0
Outcome	11	20	0
Analysis	0	4 [†]	27
Overall	0	4 [†]	27

[†] Rated “some concerns” (externally validated studies). Domain and overall applicability details are provided in Supplementary Material S9.

3.4. Data-centric techniques and deployment-readiness

Data-centric techniques were frequently reported (Table 2): label-noise/data-quality, distribution-shift, and calibration/uncertainty handling were each reported in 84% of studies, missing-modality handling in 81%, and class-imbalance handling in 74%; equity or subgroup analysis was reported in 61%. In sharp contrast, deployment-readiness indicators were rare. External validation on a separate site, dataset, or cohort was reported in only 4/31 studies (13%) [20–23]; the remainder relied on internal held-out or cross-validation. A patient, clinician, or health-system outcome was measured in only 3/31 studies (10%) [24,25], with one further framework reporting an outcome proxy [26]. Fewer than five studies cited any prediction-model reporting standard, and prospective or in-workflow deployment was not identified in any study. These high technique-reporting rates must be read in light of the search strategy, which preferentially retrieves studies that foreground such techniques, and of the coding rule: a *present* code indicates a technique was implemented or evaluated, not that robustness was rigorously demonstrated out-of-distribution.

3.5. Risk of bias

Overall risk of bias was high in 27/31 studies (87%) and of some concern in 4/31 (13%); no study was at low overall risk (Table 3, Fig. 2). The analysis domain was almost universally compromised (high in 27 studies), reflecting small or unreported sample sizes relative to model complexity, absent external validation, infrequent calibration, and limited safeguards against overfitting. The participants domain was frequently unclear (12 studies) or high (5, chiefly studies using synthetic or non-representative data), whereas the predictors domain was mostly low risk. The four studies at only “some concerns” were precisely those reporting external or temporal validation [20–23]. This 87% high-risk rate is consistent with systematic reviews of clinical machine-learning prediction models, where high risk of bias typically ranges from 84 to 100% [10], and it directly conditions the interpretation of the technique-reporting rates above.

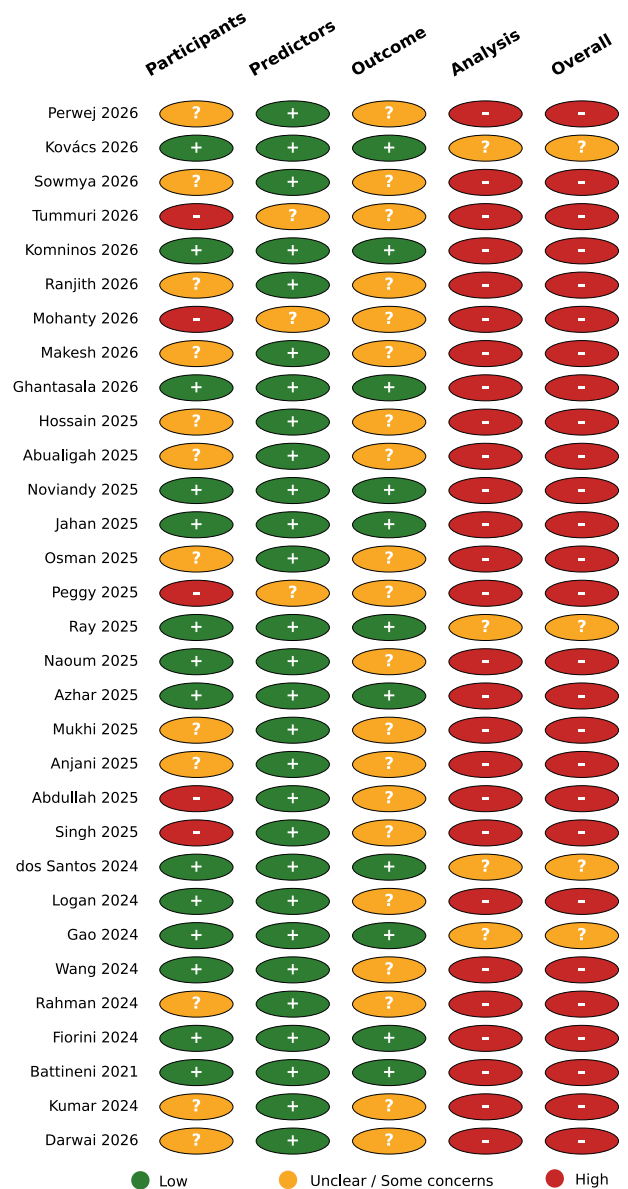


Fig. 2. PROBAST + AI risk-of-bias traffic-light plot for the 31 included studies. Green: low; amber: unclear or some concerns; red: high risk of bias. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

3.6. Relationship between fusion strategy and validation

Because developers often ask which fusion architecture yields the most clinically robust models, we cross-tabulated fusion strategy against external validation and outcome measurement. The four externally validated studies were distributed across fusion types—input/early [20], attention/gated [21], late/stacked-ensemble [22], and joint/intermediate [23]—and the three outcome-measuring studies used early, intermediate, and unspecified fusion. No fusion strategy was associated with a higher rate of external validation or outcome measurement. Based on current evidence, therefore, the choice of fusion architecture cannot be linked to clinical robustness; deployment-relevant evidence is absent across all fusion families, and this is itself the actionable finding for developers.

3.7. Sensitivity analysis

Excluding preprints and theses ($n = 5$: [20,23,27–29]) left 26 peer-reviewed studies and did not materially change the headline

proportions: external validation 3/26 (12%), measured clinical outcome 3/26 (12%), and high overall risk of bias in 23/26 (88%). The central finding – frequent technique reporting alongside near-absent external validation, outcome measurement, and high risk of bias – is therefore robust to the inclusion of non-peer-reviewed sources.

4. Discussion

4.1. Principal findings

Across 31 primary multimodal CDS studies, almost all published since 2024, data-centric robustness and explainability techniques were reported in roughly three-quarters of studies, yet external validation (13%), measured clinical outcomes (10%), and standardized reporting were nearly absent, no study reported routine deployment, and 87% were at high overall risk of bias. Within this recent literature, robustness and explainability techniques are shown to be *applicable* to multimodal CDS, not that the resulting systems are reliable or beneficial in practice. This is proof-of-concept, not proof-of-benefit.

4.2. Interpretation and scope of claims

Two boundaries must frame these results. First, because the search deliberately targets the “data-centric, robust, explainable multimodal” literature and is limited to 2021–2026, it preferentially retrieves studies that claim these properties; the reported rates describe *this recent, self-selected slice of the field*, not multimodal CDS as a whole or its full history. We therefore restrict our inferences accordingly and do not claim to characterize the entire field. Second, a robustness technique reported on a single internal dataset is a claim, not evidence: with external validation at 13% and 87% of studies at high risk of bias, the out-of-distribution reliability that “robustness” implies is largely untested *within this slice*. The juxtaposition–frequent internal technique reporting against near-absent external and outcome evidence, at high risk of bias—is the key result, and it is corroborated by the sensitivity analysis and the objective deployment-readiness indicators. This pattern mirrors adjacent clinical-AI reviews [16,17] and the broad multimodal-medicine scoping review [3], and is consistent with documented optimism and limited transportability in clinical machine-learning prediction models [7,10].

Whether “data-centric AI” represents a genuinely new paradigm or a re-branding of long-standing concerns—data quality, imbalance, missingness, calibration—is a fair question. Many constituent techniques predate the term. Our reading is that the contribution of the data-centric framing is one of emphasis and systematization rather than invention: it foregrounds data-side reliability as a first-class objective. The evidence here, however, suggests the framing currently outpaces its validation, and the label should not be taken to imply demonstrated robustness.

4.3. Comparison with existing literature

Prior reviews examined multimodal medical AI, CDS, and data-centric methods largely in isolation [16,17], with the multimodal-medicine landscape mapped separately [3]. Our review extends these to the specific intersection of data-centric robustness, explainability, and multimodality, and adds two elements not previously combined: an explicit implementation-versus-mention coding lens, and a formal PROBAST + AI appraisal that grounds the interpretation in study quality. Broader analyses of the medical literature in the AI era have likewise cautioned that reported performance frequently outpaces validated clinical utility [6]. Against deployment-oriented standards [11, 12], including formal appraisal tools [13], none of the included studies met best-practice expectations for external validation and outcome measurement.

4.4. Implications for research and practice

For researchers, continued technique development without external validation and outcome measurement will yield diminishing returns.

Four priorities follow: (1) external, multi-site validation under realistic missingness and acquisition heterogeneity; (2) measurement of at least one patient, clinician, or workflow outcome benchmarked against current practice; (3) routine adherence to prediction-model reporting and appraisal standards and formal evaluation of explanation faithfulness rather than illustrative use [11,13]; and (4) treatment of robustness claims as hypotheses to be tested out-of-distribution, not as properties established by construction. It is useful to read validation as a graded ladder—technical (internal) → temporal → external/multi-site → prospective → demonstrated clinical benefit; the studies here cluster at the first rung, with four reaching temporal or external validation and none reaching prospective or benefit. For healthcare organizations, this supports structured pilot evaluation with prospective monitoring rather than routine deployment.

4.5. Strengths and limitations

Strengths include a focused, novel question at an unaddressed intersection, an explicit coding framework with operational definitions, dual-reviewer screening and extraction, and a formal PROBAST + AI appraisal. Limitations should be acknowledged in both the evidence and the process. The evidence base is recent and self-selected by the search terms, so the technique-reporting rates reflect a specific 2021–2026 trend, and inconsistent reporting of sample size and fusion strategy limited some comparisons. For the review process, registration was retrospective; the search covered four databases without grey-literature or citation-chaining, which may have missed eligible studies; and although two reviewers screened, extracted, and rated risk of bias with consensus adjudication, formal inter-rater reliability statistics were not computed. These constraints temper the precision of individual proportions but not the central, sensitivity-tested finding.

4.6. Conclusion

In the recent data-centric multimodal CDS literature, studies widely report missing-modality handling, label-noise and distribution-shift mitigation, calibration, and explainability, yet almost none demonstrate external validity or measured clinical benefit, and the evidence is at high risk of bias. Closing the gap between technical promise and patient benefit requires external multi-site validation, outcome measurement, formal bias appraisal, and standardized reporting. Until then, and within the scope of this recent slice of the field, these systems are better regarded as promising research prototypes than as deployable decision support.

CRedit authorship contribution statement

Md Mazharul Islam: Writing – original draft, Supervision, Methodology, Formal analysis, Data curation, Conceptualization. **Abrar Mohammed Tanzim Alam:** Writing – review & editing, Validation, Methodology, Data curation. **Md. Sadikur Rahman Rony:** Writing – review & editing, Validation, Data curation. **Susmay Das:** Writing – review & editing, Validation, Data curation.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used a generative AI assistant to edit language and formatting. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data for this article can be found online at doi:10.1016/j.ijmedinf.2026.106684.

Data availability

The extraction dataset, screening decisions, full-text exclusion list, search strings, and risk-of-bias assessments are provided as Supplementary Material.

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